Impulse changes momentum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect force over time to momentum change.

SUCCESS CRITERIA - I can...

1. Calculate impulse from force and duration.
2. Match impulse direction to momentum change.
3. Use impulse to find a final velocity.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A planet has surface field **75 N/kg**. At center distance **5** R, find field and the force on a **7 kg** object.

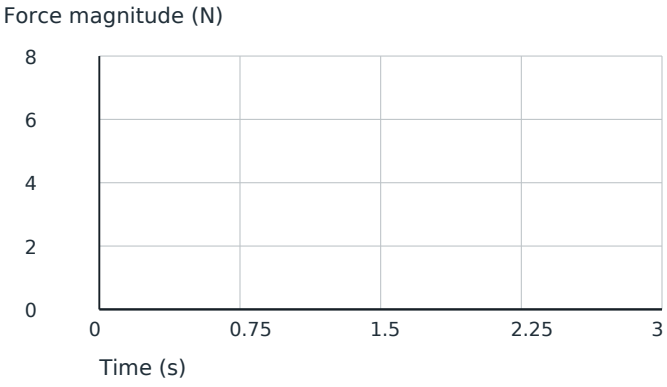
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $J = F_{avg}\,\Delta t = \Delta p = p_f - p_i$.
J is impulse; F_{avg} is average net force. Signed force-time area gives impulse; i = initial, f = final.

Worked reasoning: A constant **6 N** force acts right for **2 s**. Impulse is **12 N·s** right, so momentum increases by **12 kg·m/s**.

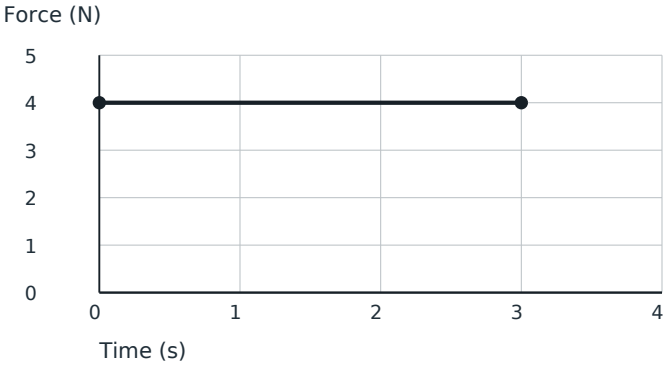
Your turn: 1. **Draw the force-time rectangle** and **shade its area**.
2. **Explain why the two impulse** units are equivalent.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Read the rectangular force-time graph**.
2. **Find the area** and **state what it means**.



Marzano check - circle: ① ② ③

04 Apply the physics

A **2 kg** cart starts at rest and receives **6 N·s** rightward impulse. Final momentum is **6 kg·m/s**. **Find final velocity** by dividing by mass.

Before	After

Label mass and velocity.

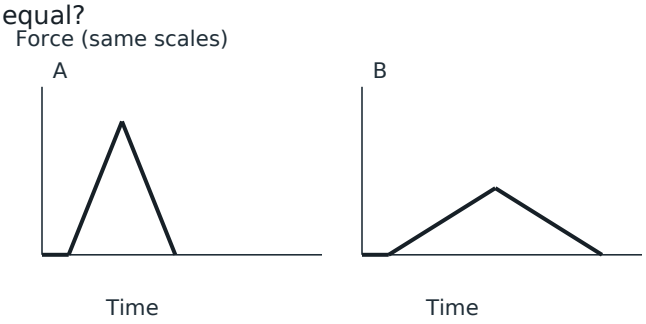
Marzano check - circle: ① ② ③

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05 Explain your reasoning

Two force pulses have equal signed area but different peak heights. What must be equal? What need not be equal?

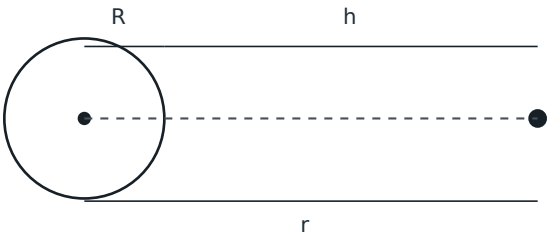


Same scales: compare impulse and peak force.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

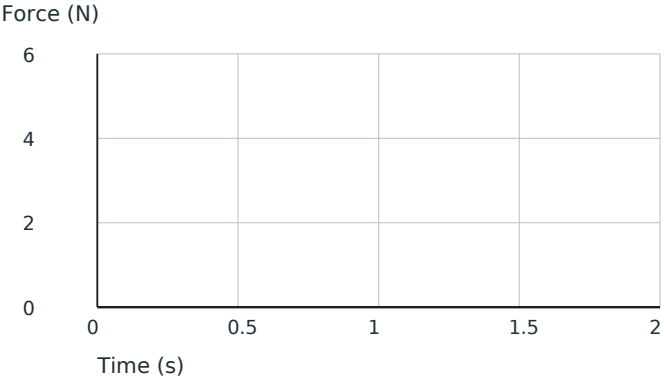
A spherical planet has surface field **144 N/kg**. **Find the field** at center distance **6** R, where R is the planet radius. **Label the masses**, field or force relevant to the question.



Marzano check - circle: 1 2 3

07 Exit check

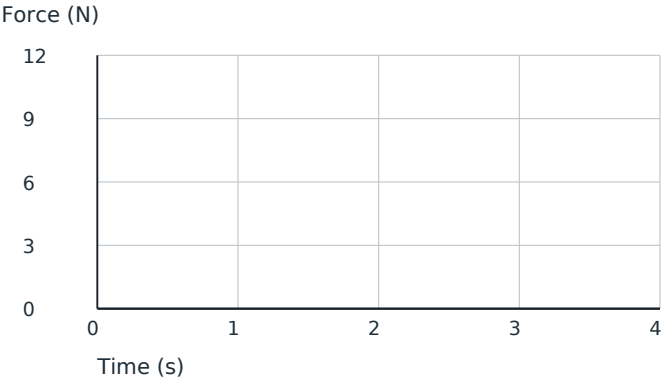
A **5 N** force acts right for **2 s**. **Find impulse**. On a **2 kg** cart initially at rest, find final velocity.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. Create two constant force-and-time pairs giving an impulse of **12 N·s**.
2. **Compare their force-time rectangles**.



Marzano check - circle: 1 2 3